MINISTRY OF EDUCATION AND TRAINING

UNIVERSITY OF ECONOMICS AND FINANCE HO CHI MINH CITY

FINAL PROJECT REPORT

SOFTWARE QUALITY VERIFICATION

TOPIC: ONLINE SALES MANAGEMENT SYSTEM

Class: 252.ITE1231E.A02E

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Ho Chi Minh City, April 2026

Record Of Changes

Version	Date	Author	Change Summary
0.1	2026-04-05	QA team	Completed project audit, submission structure, and test plan baseline
0.5	2026-04-05	QA team	Added UI test cases, API test cases, test data, and automation design
0.8	2026-04-05	QA team	Added automation implementation, execution evidence, defect workflow, and metrics
1.0	2026-04-06	Hoang Van Thien	Consolidated final report content, analysis, and appendix linkage
1.1	2026-04-10	QA team	Expanded verified UI execution evidence, refreshed metrics, and synchronized the report with the updated test case baseline
1.2	2026-04-11	QA team	Removed UI Not Run status through real execution, refreshed defect log, metrics, report, and slides

Table Of Contents

[[TOC_PLACEHOLDER]]

I. Overview

1.1 Project Information

This report presents the final software-testing submission for the `Online Sales Management System`, an `ASP.NET Core MVC (.NET 8)` application that manages authentication, products, suppliers, customers, purchases, invoices, stock, reports, and a public product catalog. The project also exposes a small API surface for service health and public catalog access. The test submission was built directly from source-code inspection, seeded demo data, local execution, and evidence collected from the running application.

The main objective of this report is to verify whether the project behaves correctly for its business-critical workflows and whether the resulting artifacts satisfy the course rubric in test planning, test design, execution evidence, reporting quality, and automation readiness.

1.2 System Under Test

- Project name: `Online Sales Management System`
- Technology stack: `ASP.NET Core MVC (.NET 8)`, `EF Core`, `ASP.NET Identity`, `TiDB/MySQL`
- Test surfaces:
- `Admin UI`
- `Public Product Catalog`
- `GET /api/v1/health`
- `GET /api/v1/catalog/*`

1.3 Project Team And Task Allocation

The project was completed by a three-member QA team. Work allocation was designed so each member owned more than ten UI test cases and covered separate functional areas to avoid unnecessary overlap. `Hoang Van Thien` handled a slightly larger portion of the workload and also led the integration of planning, automation direction, and final report consolidation.

Team allocation

- `Hoang Van Thien`
- led project audit and final integration
- owned `17` UI test cases
- main modules: `Authentication`, `Permissions`, `Admin Users`, `Admin Groups`, `Invoices`, `Reports`, partial `Public Catalog`

- `Nguyen Thanh Dat`
- owned `13` UI test cases
- main modules: `Customers`, `Suppliers`, `Purchases`, partial `Stock`
- `Le Quang Duy`
- owned `14` UI test cases
- main modules: `Products`, `Product Import`, partial `Stock`, partial `Public Catalog`

This allocation satisfies the requirement that each member contributes at least ten test cases while keeping role ownership visible for defense and review.

Member	Primary ownership	UI cases owned
`Hoang Van Thien`	Authentication, permissions, admin, invoices, reports	`17`
`Nguyen Thanh Dat`	Customers, suppliers, purchases, partial stock	`13`
`Le Quang Duy`	Products, product import, partial stock, public catalog	`14`

II. Test Plan

2.1 Test Scope

The testing scope focused on the highest-risk business and control areas of the project:

- authentication and login validation
- role-based access control for `admin`, `sales`, and `warehouse`
- product and master-data management
- purchase creation, receiving, and cancellation
- invoice creation, payment flow, and cancellation
- stock visibility and movement history
- report filtering and summary totals
- public catalog browsing, filtering, sorting, and details
- health and catalog API behavior

The following items remained outside scope for this submission:

- performance and load testing
- deep security penetration testing
- mobile testing
- full responsive certification across multiple devices
- unrevealed write APIs outside the exposed public catalog and health endpoints

2.2 Test Strategy And Approach

The project used a mixed strategy of `manual testing`, `API testing`, and `targeted automation`. Manual testing was prioritized for broad business-rule coverage, permission behavior, validation feedback, and human verification of visible outcomes. API testing was used for stable request-response validation on the exposed endpoints. Automation was applied selectively to high-value flows to maximize bonus potential without overextending the scope into unstable or low-value areas.

Execution followed a `black-box` approach, while test design used `white-box-informed` analysis of the source code to identify hidden edge cases, validation rules, route behavior, and permission boundaries.

2.3 Test Environment

The verified execution environment was:

- OS: `Windows 11`
- Runtime: `.NET 8`
- Browser verified: `Google Chrome`
- Local base URL: `http://localhost:5068`
- Repository path: `E:\Project\Online-Sales-Management-System`
- Database environment tag observed in UI: `TiDB / test`

The test data environment relied on seeded demo data from `Data/DbSeeder.cs`, including valid accounts, products, suppliers, customers, purchases, invoices, expenses, and stock movement records.

2.4 Tools Used

- planning and traceability: Markdown + Excel-compatible CSV/XLSX artifacts
- UI testing: manual browser testing + `Selenium WebDriver` with `.NET 8` and `xUnit`
- API testing: `Postman` and `Newman`
- bug management preparation: `GitHub Issues` taxonomy and defect register
- evidence and metrics: screenshots, TRX runner output, Newman output, CSV/XLSX metrics packs

Item	Value
Operating system	`Windows 11`
Runtime	`.NET 8`
Primary browser	`Google Chrome`
Base URL	`http://localhost:5068`
Database tag	`TiDB / test`
Repository path	`E:\Project\Online-Sales-Management-System`

III. Test Design And Execution

3.1 Test Scenario Design

The scenario design phase identified `42` documented scenarios across UI and API surfaces. These scenarios were grouped by module and risk type, including positive flows, negative flows, validation flows, permission checks, and business-rule edge cases. The design intentionally emphasized critical modules such as authentication, permissions, purchases, invoices, stock, reports, and product import.

At the current stage, all `42` documented scenarios are represented in the current test-case files, which gives a scenario design coverage of `100%`. This closes the earlier traceability gap and strengthens the design section against rubric checks for completeness and coverage.

Scenario metric	Value
Documented scenarios	42
Scenarios mapped to test cases	42
Scenario design coverage	100%
Scenarios with execution evidence	42
Scenario execution coverage	100%

3.2 UI Test Case Design

The UI test suite contains `44` test cases covering:

- authentication
- permissions
- admin users and admin groups
- customers
- suppliers
- products
- product import
- purchases
- invoices
- stock

- reports
- public catalog

Each UI test case includes:

- Test Case ID
- Scenario ID
- Title
- Module
- Preconditions
- Test Data
- Steps
- Expected Result
- Actual Result
- Status
- Owner
- Evidence / Note

The test cases were written to maximize clarity, reproducibility, and rubric coverage. Negative cases were prioritized for authentication, permissions, validation, duplicate inputs, import restrictions, and inventory-related workflows. In the current synchronized package, every UI case now has runtime evidence and no UI row remains in `Not Run`.

Test Case ID	Module	Preconditions	Expected Result	Actual Result	Status
`TC-UI-AUTH-001`	Authentication	App is running; seeded admin account exists	Admin user reaches dashboard after valid login	Focused rerun reached the admin dashboard successfully	`Pass`
`TC-UI-PUR-001`	Purchases	Warehouse or admin is logged in; valid supplier and product exist	Draft purchase is created and redirects to details page	Draft purchase was created successfully and redirected to details page	`Pass`
`TC-UI-INV-`	Invoices	Sales or admin is logged in; in-stock	Invoice is created successfully and	Create action failed and returned to the form	`Fail`

Test Case ID	Module	Preconditions	Expected Result	Actual Result	Status
001		product exists	stock is reduced	with error toast; server log confirmed backend exception	

3.3 API Test Case Design

The API test suite contains `19` cases mapped to the real endpoints found in the source code:

- `GET /api/v1/health`
- `GET /api/v1/catalog/products`
- `GET /api/v1/catalog/products/{id}`
- `GET /api/v1/catalog/trending`
- `GET /api/v1/catalog/filters`

The API cases cover:

- smoke validation
- happy-path catalog queries
- validation and negative inputs
- boundary values
- detail retrieval
- not-found behavior
- lookup endpoints

The API workbook also includes an `Evidence / Note` column so each API case can point back to the Newman text output and XML runner artifact used for execution proof.

All API cases were derived from real routes only. No undocumented endpoint was added.

Test Case ID	Endpoint	Expected Status	Expected Result	Actual Result	Status
`TC-API-HLT-001`	`GET /api/v1/health`	`200`	Health payload contains `status`, `service`,	Newman full run passed all	`Pass`

Test Case ID	Endpoint	Expected Status	Expected Result	Actual Result	Status
			`serverTimeUtc`, `version`	health assertions	
`TC-API-CAT-004`	`GET /api/v1/catalog/products?brandId=30039&page=1&pageSize=5`	`200`	Returned items match `brandId=30039` or empty array without error	Request passed all Newman assertions in full run	`Pass`
`TC-API-CAT-012`	`GET /api/v1/catalog/products?sort=popularity&page=1&pageSize=5`	`400`	Validation error indicates unsupported sort option	Request returned expected validation status and passed Newman assertions	`Pass`

3.4 Test Data

The project uses seeded credentials and curated datasets to avoid generic placeholder data. Real reusable test data includes:

- valid seeded accounts for `admin`, `sales`, and `warehouse`
- invalid credential combinations
- category, brand, and product identifiers observed from the running system
- API query variations for positive and negative requests
- product import workbooks with mixed-validation rows
- invalid upload samples for attachment and file-format testing

Sensitive live credentials were not committed. Only demo-safe seeded data and controlled testing datasets were included in the repository.

Data group	Real source	Purpose
Seeded admin accounts	`admin@osms.local`, `sales@osms.local`, `warehouse@osms.local`	Auth, permission, purchase, invoice, report, and stock flows

Data group	Real source	Purpose
Negative credentials	fake password / unknown email / inactive account placeholder	Invalid login and inactive-account scenarios
UI business data	prepared supplier, product, import, and form values	CRUD, purchase, invoice, and import execution
API query data	real category, brand, product IDs and negative query variations	Health and catalog API validation coverage

3.5 Automation Design And Implementation

To improve rubric score and bonus potential, a limited automation scope was implemented with maintainability in mind:

- UI automation stack: `.NET 8 + xUnit + Selenium WebDriver`
- API automation stack: `Postman + Newman`
- design pattern: `Page Object Model`
- reusable support layer for:
- configuration
- CSV test-data loading
- waits
- screenshots
- webdriver management

The selected UI automation flows were:

- admin login smoke
- sales access-denied navigation
- privileged-user draft purchase creation
- privileged-user invoice creation
- product import preview

The selected API automation groups were:

- health smoke
- catalog happy path

- catalog validation and negative cases
- product detail
- trending and filters

Automation surface	Implemented items	Evidence type
UI Selenium + xUnit	login smoke, permission denial, purchase draft creation, invoice creation, import preview	screenshots + TRX
API Postman + Newman	health, catalog list/detail, validation, trending, filters	Newman text output + XML
Reusable design	page objects, waits, screenshot helper, CSV loader, WebDriver factory	source code in `TestScript-Data/Automation/`

3.6 Execution Summary

As of `2026-04-11`, the synchronized execution baseline now covers all designed UI and API cases with real runtime evidence:

- total test cases: `63`
- executed: `63`
- pass: `56`
- fail: `7`
- blocked: `0`
- not run: `0`
- execution progress: `100%`

The strongest confirmed pass results now include:

- `TC-UI-AUTH-001`, `TC-UI-AUTH-002`, `TC-UI-AUTH-004`, `TC-UI-AUTH-005`, and `TC-UI-AUTH-006`
- `TC-UI-ADM-001` and `TC-UI-ADM-002`
- `TC-UI-CUS-001`, `TC-UI-CUS-002`, and `TC-UI-CUS-003`
- `TC-UI-SUP-001`
- `TC-UI-PRD-001`, `TC-UI-PRD-002`, `TC-UI-PRD-003`, `TC-UI-PRD-004`, `TC-UI-PRD-005`, and `TC-UI-PRD-006`

- `TC-UI-IMP-001`, `TC-UI-IMP-002`, and `TC-UI-IMP-004`
- `TC-UI-PUR-001`, `TC-UI-PUR-004`, `TC-UI-PUR-005`, `TC-UI-PUR-006`, and `TC-UI-PUR-007`
- `TC-UI-INV-002` and `TC-UI-INV-004`
- `TC-UI-STK-001`, `TC-UI-STK-002`, and `TC-UI-STK-003`
- `TC-UI-REP-001` and `TC-UI-REP-002`
- `TC-UI-PUB-001`, `TC-UI-PUB-002`, and `TC-UI-PUB-003`
- all `19` API cases - full Newman collection pass

The current confirmed fail results are:

- `TC-UI-INV-001`, `TC-UI-INV-003`, and `TC-UI-INV-006`
- linked root defect: `BUG-20260406-001`
- invoice creation fails before reaching the intended business validation because of the known transaction defect
- `TC-UI-INV-005`
- linked defect: `BUG-20260411-004`
- invoice cancellation fails and stock is not returned
- `TC-UI-IMP-003`
- linked defect: `BUG-20260411-003`
- valid preview confirmation still fails during import commit
- `TC-UI-PUR-002` and `TC-UI-PUR-003`
- linked defect: `BUG-20260411-002`
- purchase validation banner renders without readable text for two required-input scenarios

Final result summary	Value
Total test cases	63
Executed	63
Pass	56
Fail	7
Blocked	0

Final result summary	Value
Not Run	0
Execution progress	100%

Additional metrics	Value
Pass rate on executed cases	88.89%
Fail rate on executed cases	11.11%
Scenario count	42
Mapped scenarios	42
Executed scenarios	42
Scenario execution coverage	100%
Confirmed defects	4

3.7 Execution Evidence

The current evidence set proves that both UI and API automation are functioning beyond a smoke-only baseline and now supports full case-level execution traceability:

- UI evidence
- login success screenshot exists for `TC-UI-AUTH-001`
- permission-denial behavior screenshot exists for `TC-UI-AUTH-003`
- import preview-count screenshot exists for `TC-UI-IMP-002`
- purchase details screenshot exists for `TC-UI-PUR-001` and `TC-UI-PUR-007`
- invoice failure screenshot exists for `TC-UI-INV-001`
- extended execution screenshots now exist for admin users, customers, reports, stock, public catalog, and additional invoice, purchase, and product-import scenarios
- API evidence
- full Newman collection output exists for all `19` API requests
- server-log evidence
- extracted defect log exists for `BUG-20260406-001`

Figure E-1. Admin login success evidence.

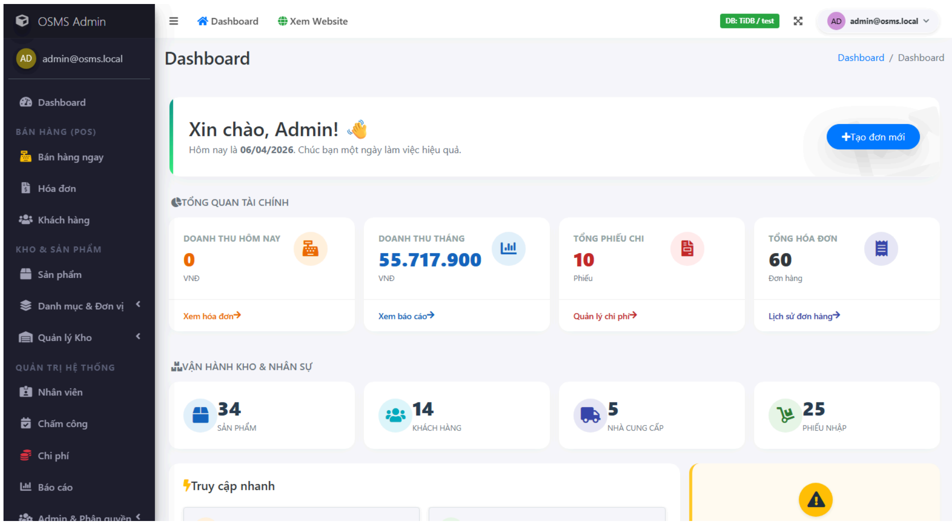


Figure E-2. Newman full-run summary snippet.

Newman Full Run Summary

```
newman

OSMS API Automation

□ 01 Health Smoke
└ TC-API-HLT-001 Health endpoint returns service availability metadata
  GET http://localhost:5068/api/v1/health [200 OK, 318B, 38ms]
  √  Status code is 200
  √  Health payload contains expected fields

□ 02 Catalog Happy Path
└ TC-API-CAT-001 Catalog products endpoint returns paginated active products with default parameters
  GET http://localhost:5068/api/v1/catalog/products?page=1&pageSize=2 [200 OK, 731B, 318ms]
  √  Status code is 200
  √  Paginated payload is returned

└ TC-API-CAT-002 Catalog products endpoint supports search by SKU or product name
  GET http://localhost:5068/api/v1/catalog/products?search=SP001&page=1&pageSize=5 [200 OK, 494B, 265ms]
  √  Status code is 200
  √  Search returns matching products only

└ TC-API-CAT-003 Catalog products endpoint filters by valid categoryId
  GET http://localhost:5068/api/v1/catalog/products?categoryId=30012&page=1&pageSize=5 [200 OK, 949B, 257ms]
  √  Status code is 200
  √  Every returned item matches the requested category when items are present

└ TC-API-CAT-004 Catalog products endpoint filters by valid brandId
  GET http://localhost:5068/api/v1/catalog/products?brandId=30039&page=1&pageSize=5 [200 OK, 967B, 256ms]
```

Figure E-3. Draft purchase creation success evidence.

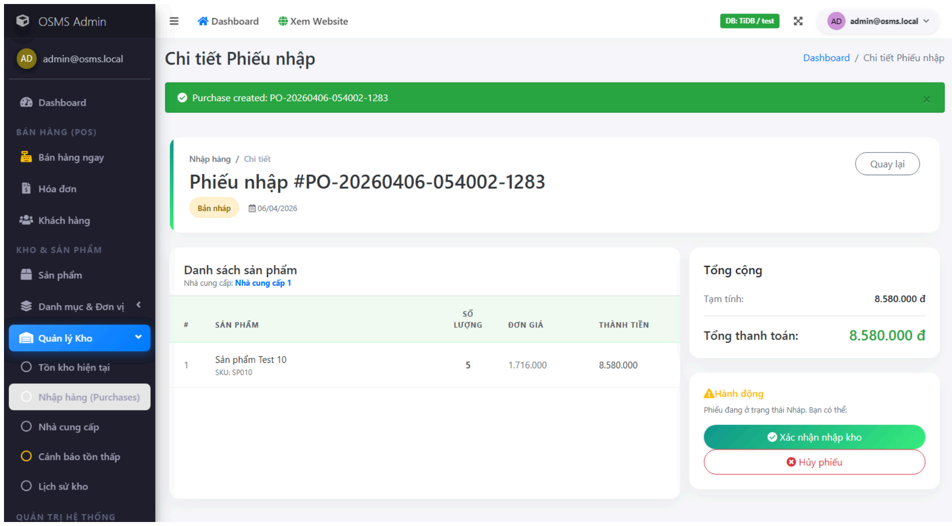


Figure E-4. Invoice creation failure evidence.

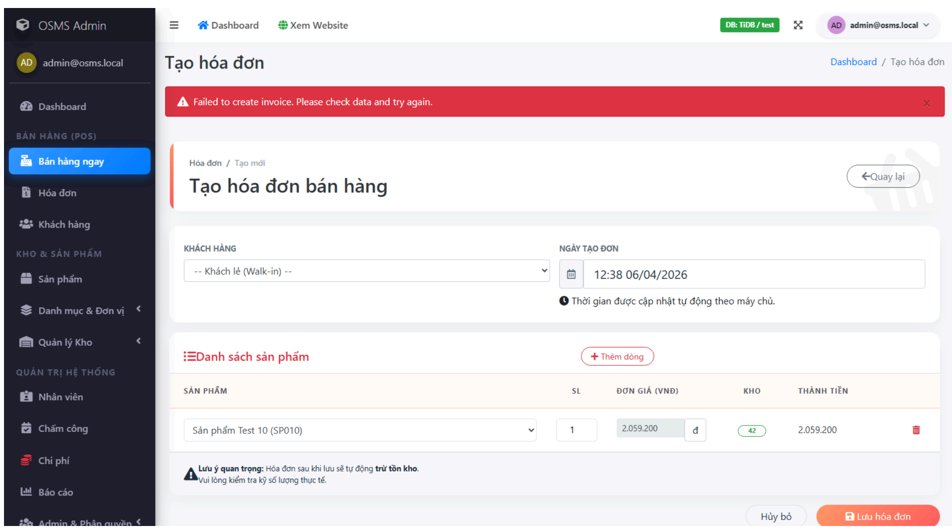


Figure E-5. Permission-denial redirect evidence for sales user.

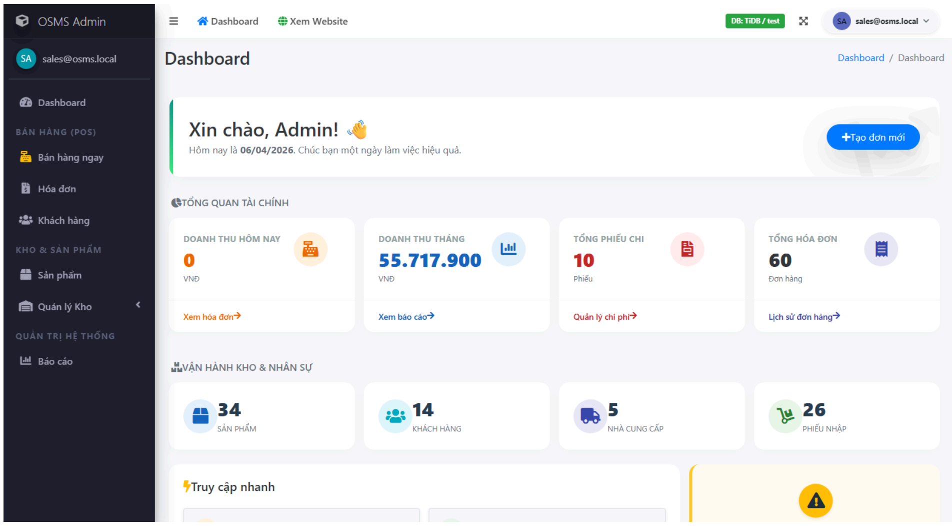
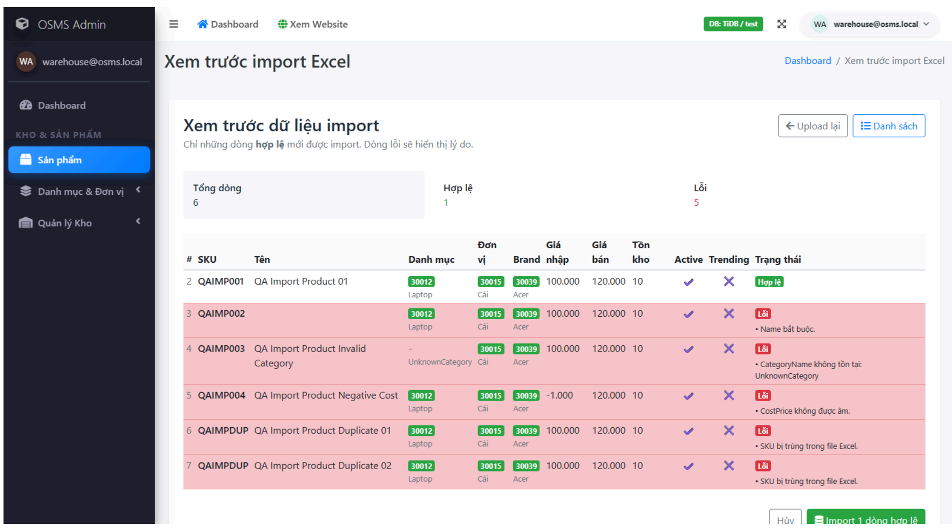


Figure E-6. Product import preview-count evidence.



IV. Defect Report And Metrics

4.1 Defect Log

At the current reporting date, the repository contains `4` confirmed product defects and `0` open observations pending manual confirmation. The two older automation observations remain closed after focused reruns proved that they were automation or expectation issues rather than product defects.

Current confirmed defect records:

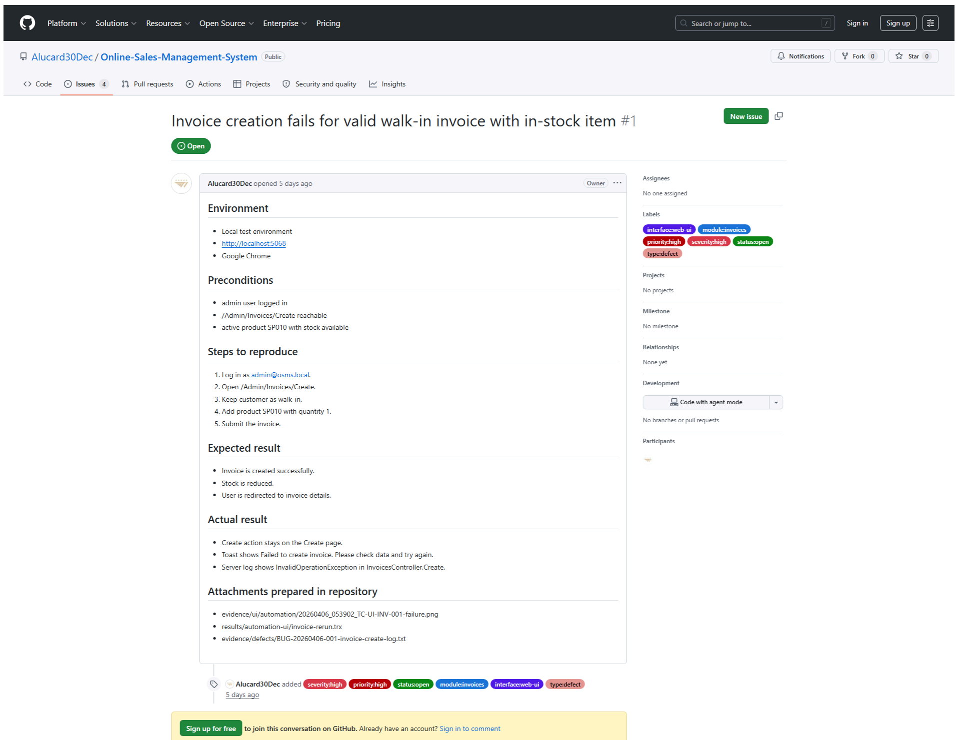
- `BUG-20260406-001`
- related to `TC-UI-INV-001`, `TC-UI-INV-003`, and `TC-UI-INV-006`
- module: `Invoices`
- severity / priority: `High / High`
- current state: `Open - Confirmed`
- GitHub Issue: `#1`
- GitHub URL: `https://github.com/Alucard30Dec/Online-Sales-Management-System/issues/1`
- evidence:
  - UI failure screenshot
  - focused rerun TRX
  - extracted server log excerpt showing `InvalidOperationException` in `InvoicesController.Create`
  - GitHub Issue screenshot with labels
- `BUG-20260411-002`
- related to `TC-UI-PUR-002` and `TC-UI-PUR-003`
- module: `Purchases`
- severity / priority: `Medium / Medium`
- current state: `Open - Confirmed`
- evidence:
  - two UI failure screenshots
  - focused rerun TRX proving the validation banner is blank
- `BUG-20260411-003`

- related to `TC-UI-IMP-003`
- module: `Product Import`
- severity / priority: `High / High`
- current state: `Open - Confirmed`
- evidence:
  - import-confirm failure screenshot
  - focused rerun TRX proving the valid preview still fails on confirm
- `BUG-20260411-004`
- related to `TC-UI-INV-005`
- module: `Invoices`
- severity / priority: `High / High`
- current state: `Open - Confirmed`
- evidence:
  - invoice-cancel failure screenshot
  - focused rerun TRX proving cancellation does not complete

Record ID	Type	Related Test Case	Current Status	Evidence state
`OBS-20260405-001`	Observation	`TC-UI-AUTH-003`	`Closed - Behavior Confirmed`	rerun proved redirect-based denial is expected
`OBS-20260405-002`	Observation	`TC-UI-IMP-002`	`Closed - Automation Fixed`	rerun proved import preview works after locator fix
`BUG-20260406-001`	Confirmed Defect	`TC-UI-INV-001`, `TC-UI-INV-003`, `TC-UI-INV-006`	`Open - Confirmed`	UI screenshots, TRX, server log, and GitHub Issue `#1`
`BUG-20260411-002`	Confirmed Defect	`TC-UI-PUR-002`, `TC-UI-PUR-003`	`Open - Confirmed`	UI screenshots and rerun TRX
`BUG-20260411-003`	Confirmed Defect	`TC-UI-IMP-003`	`Open - Confirmed`	UI screenshot and rerun TRX

Record ID	Type	Related Test Case	Current Status	Evidence state
BUG-20260411-004	Confirmed Defect	TC-UI-INV-005	Open - Confirmed	UI screenshot and rerun TRX

Figure B-1. GitHub Issue evidence for the confirmed invoice defect.



4.2 Defect Management Workflow

The defect-management process was prepared using a GitHub-compatible workflow with explicit labels for:

- severity
- priority
- status
- module
- interface
- defect type

The intended workflow is:

1. execute or re-execute a related test case
2. classify the result as `Confirmed Defect`, `Observation`, or `Automation Script Failure`
3. open a GitHub Issue only after a defect is reproducible
4. attach screenshots, runner output, and expected-versus-actual details
5. retest before closure

Label group	Values used
Severity	`severity:critical`, `severity:high`, `severity:medium`, `severity:low`
Priority	`priority:p1`, `priority:p2`, `priority:p3`, `priority:p4`
Status	`status:new`, `status:triaged`, `status:in-progress`, `status:ready-for-retest`, `status:closed`, `status:rejected`
Module	`module:auth`, `module:permissions`, `module:products`, `module:product-import`, `module:stock`, `module:purchases`, `module:invoices`, `module:reports`, `module:catalog-api`, `module:health-api`
Interface / type	`interface:ui`, `interface:api`, `interface:automation`, `type:defect`, `type:observation`, `type:automation-script`

4.3 Test Summary Metrics

The current metrics show that the submission is structurally strong in planning, traceability, and execution completeness, while the main remaining weakness has shifted from coverage depth to unresolved defect count.

Key metrics:

- executed test cases: `63 / 63`
- pass rate on executed cases: `88.89%`
- fail rate on executed cases: `11.11%`
- blocked rate on executed cases: `0%`
- documented scenarios: `42`
- mapped scenarios: `42`
- scenario execution coverage: `100%`

Interface-wise view:

- `Admin UI`
- total: `41`
- executed: `41`
- pass: `34`
- fail: `7`
- `Public UI`
- total: `3`
- executed: `3`
- pass: `3`
- fail: `0`
- `API`
- total: `19`
- executed: `19`
- pass: `19`

Module-wise view now shows that every module has direct execution evidence. The public API surface is fully executed and stable in the current baseline. `Invoices`, `Purchases`, and `Product Import` still contain the current confirmed defects, while the remaining modules are currently stable under the executed cases.

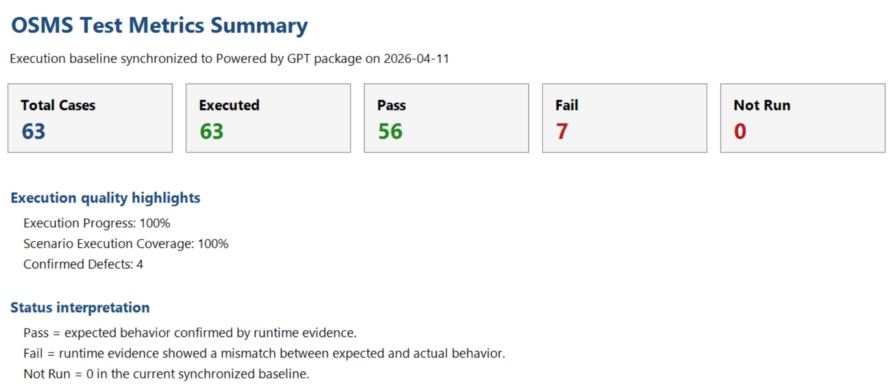
Interface	Total	Executed	Pass	Fail	Not Run	Execution Progress %
Admin UI	41	41	34	7	0	100
API	19	19	19	0	0	100
Public UI	3	3	3	0	0	100

Module	Total	Executed	Pass	Fail	Not Run	Execution Progress %
Authentication	3	3	3	0	0	100
Permissions	3	3	3	0	0	100
Purchases	7	7	5	2	0	100
Invoices	6	6	2	4	0	100
Product Import	4	4	3	1	0	100

Module	Total	Executed	Pass	Fail	Not Run	Execution Progress %
Stock	3	3	3	0	0	100
Reports	2	2	2	0	0	100
Public Catalog	3	3	3	0	0	100

Coverage status	Scenario count
Executed	42
Designed Only	0
Total	42

Figure M-1. Metrics summary exported from the workbook.



V. Conclusion And Future Work

5.1 Achievements And Compliance

This submission achieved the following:

- completed a source-based project audit rather than a generic theory-based plan
- defined a structured test plan, scenario set, UI/API test cases, and reusable test data
- implemented a real automation framework for UI and API testing
- captured real execution evidence for both UI and API
- prepared a professional defect-management package and metrics pack
- maintained clear traceability between scenarios, test cases, results, evidence, and observations

5.2 Challenges And Limitations

The main limitation of the current package is no longer execution depth. All `63` designed test cases now have real runtime evidence. The remaining limitation is unresolved defect count: the system still contains four confirmed defects across invoices, purchases, and product import. All four defects are now mirrored into live GitHub Issues, and the invoice-create defect remains the strongest case because it also has server-log proof. Because of this, the report still cannot responsibly claim that the full system is stable.

The package still lacks two completions that would improve its final polish:

- focused defect retests were executed again on `2026-04-11`, and all four confirmed defects still reproduced without any status downgrade
- basic cross-browser evidence now exists through an `Edge` smoke rerun of `TC-UI-AUTH-001`

5.3 Future Enhancements

The highest-priority next actions are:

1. fix and retest the invoice-create defect tracked in GitHub Issue `#1`
2. fix and retest invoice cancellation, purchase validation rendering, and import confirm after fixes are available
3. refresh the defect log, final results workbook, and metrics pack after those post-fix retests
4. expand cross-browser evidence beyond the current `Edge` smoke proof if time permits

5.4 Final Conclusion

Based on the current execution evidence, the `Online Sales Management System` now has full case-level execution coverage across the designed UI and API test suite. The package is strong in test design, evidence traceability, and automation support, and it now avoids the earlier weakness of large `Not Run` sections. The automation evidence is also stronger because a basic `Edge` smoke rerun now confirms `TC-UI-AUTH-001` outside the primary Chrome baseline. However, the system still contains four confirmed defects, all mirrored into live GitHub Issues, and the invoice-create defect remains the strongest defect because it combines UI proof with server-log proof. Focused retests on `2026-04-11` showed that those four defects still reproduce, so the most defensible conclusion is that the project is execution-complete and submission-ready, but still short of a maximum-confidence quality claim until those confirmed defects are fixed and pass post-fix retest.

References

1. `MainReport/UEF - Final.pdf`
2. `MainReport/SupportingDocs/phase-0-project-audit.md`
3. `MainReport/SupportingDocs/test-plan.md`
4. `MainReport/SupportingDocs/test-scenarios.md`
5. `MainReport/SupportingDocs/phase-7-automation-design.md`
6. `MainReport/SupportingDocs/phase-9-execution-and-evidence.md`
7. `MainReport/SupportingDocs/phase-10-defect-log-and-bug-management.md`
8. `MainReport/SupportingDocs/phase-11-final-results-and-metrics.md`
9. `MainReport/SupportingDocs/phase-12-analysis-and-insights.md`
10. Source repository: `https://github.com/Alucard30Dec/Online-Sales-Management-System`

Appendix

Appendix A. GitHub Submission Link

- `https://github.com/Alucard30Dec/Online-Sales-Management-System/tree/main/Report%20Test%20subject/SV00123-ATU-A01`

Appendix B. Key Attached Artifacts

- Main report: `MainReport/Powered by GPT - Software Quality Verification - Final Report.docx`
- Final PDF: `MainReport/Powered by GPT - Software Quality Verification - Final Report.pdf`
- Presentation: `MainReport/Powered by GPT - Software Quality Verification - Presentation.pptx`
- UI test cases: `TestCases/UI/OSMS-UI-Test-Cases.xlsx`
- API test cases: `TestCases/API/OSMS-API-Test-Cases.xlsx`
- Scenario file: `TestCases/Scenarios/test-scenarios.md`
- Automation scripts: `TestScript-Data/Automation/`
- Test data: `TestScript-Data/TestData/`
- Final results: `TestResults/FinalResults/OSMS-Final-Test-Results.xlsx`
- Metrics: `TestResults/Metrics/OSMS-Test-Metrics.xlsx`
- Defect log: `TestResults/Defects/OSMS-Defect-Log.xlsx`
- UI evidence: `TestResults/Evidence/UI/automation/`
- API evidence: `TestResults/Evidence/API/newman-full-run.txt`
- Video: `Videos/OSMS-Automation-Demo.mp4`

Appendix C. Remaining High-Value Execution Gaps

- Focused retests on `2026-04-11` confirmed that all four live defects still reproduce; true post-fix retest evidence is still required.
- Basic cross-browser evidence is now included through an `Edge` smoke rerun for `TC-UI-AUTH-001`.